STATISTICS 652

LECTURE 11 – 2026

SAMPLING DISTRIBUTION

THEOREM 11.1. FOR $\underset{\sim}{X}_1, \ldots, \underset{\sim}{X}_n$ i.i.d $N_q(\underset{\sim}{\mu}, \Sigma)$, $\Sigma \geq 0$

i) $\bar{\underset{\sim}{X}} \sim N_q(\underset{\sim}{\mu}_{\bar{X}}, \Sigma_{\bar{X}})$ with $\underset{\sim}{\mu}_{\bar{X}} = \underset{\sim}{\mu}$ and $\Sigma_{\bar{X}} = \frac{1}{n}\Sigma$

ii) $\bar{\underset{\sim}{X}}$ and S are INDEPENDENT

PROOF: i) FOLLOWS FROM ADDITIVITY OF NORMAL (LEMMA 11.5) AND CLOSURE UNDER scalar multiplication.

ii) APPROACH I. Let $\underset{\sim}{r} = \begin{pmatrix} \underset{\sim}{r}_1 \\ \vdots \\ \underset{\sim}{r}_n \end{pmatrix}$ where $\underset{\sim}{r}_i = \underset{\sim}{X}_i - \bar{\underset{\sim}{X}}$, then

$\begin{pmatrix} \underset{\sim}{r} \\ \bar{\underset{\sim}{X}} \end{pmatrix} \sim N_{(n+1)q}$ SINCE IT IS A LINEAR COMBINATION OF $\begin{pmatrix} \underset{\sim}{X}_1 \\ \vdots \\ \underset{\sim}{X}_n \end{pmatrix} \sim N_{nq}$

NOTE: $\begin{pmatrix} \underset{\sim}{r} \\ \bar{\underset{\sim}{X}} \end{pmatrix}$ has a SINGULAR $N_{(n+1)q}$ distribution

NOW $\mathrm{COV}(\underset{\sim}{r}_i, \bar{\underset{\sim}{X}}) = \mathrm{COV}(\underset{\sim}{X}_i, \bar{\underset{\sim}{X}}) - \mathrm{COV}(\bar{\underset{\sim}{X}}, \bar{\underset{\sim}{X}}) = \frac{1}{n}\sum_{j=1}^{n} \mathrm{COV}(\underset{\sim}{X}_i, \underset{\sim}{X}_j) - \mathrm{VAR}(\bar{\underset{\sim}{X}})$

$= \frac{1}{n}\mathrm{VAR}(\underset{\sim}{X}_i) - \mathrm{VAR}(\bar{X}) = \frac{1}{n}\Sigma - \frac{1}{n}\Sigma = 0.$

$\Rightarrow \Sigma_{\underset{\sim}{r}, \bar{\underset{\sim}{X}}} \equiv \mathrm{COV}(\underset{\sim}{r}, \bar{\underset{\sim}{X}}) = 0 \Rightarrow \underset{\sim}{r}$ and $\bar{\underset{\sim}{X}}$ are independent

$\Rightarrow \bar{\underset{\sim}{X}}$ and S ARE INDEPENDENT SINCE $S = \frac{1}{n-1}\sum_{i=1}^{n} \underset{\sim}{r}_i \underset{\sim}{r}_i'$.

RECALL: FUNCTIONS OF INDEPENDENT RANDOM VECTORS ARE INDEPENDENT.

APPROACH 2: By using Basu's Theorem,

For fixed Σ, $\bar{\underset{\sim}{X}}$ is complete and minimal sufficient, and

$\underset{\sim}{r}$ is ancillary $\Rightarrow$ $\bar{\underset{\sim}{X}}$ and $\underset{\sim}{r}$ independent

NOTE: $\underset{\sim}{r}_i = \underset{\sim}{X}_i - \bar{\underset{\sim}{X}} = \underset{\sim}{Y}_i - \bar{\underset{\sim}{Y}}$ where $\underset{\sim}{Y}_i = \underset{\sim}{X}_i - \underset{\sim}{\mu} \sim N_q(0, \Sigma)$, and

hence distribution of $\underset{\sim}{r}$ does not depend on $\underset{\sim}{\mu}$. □

- To complete sampling distribution of $(\bar{\underset{\sim}{X}}, S)$,

ii) $(n-1)S = n S_n \sim$ Wishart $(n-1, \Sigma)$ Advanced multivariate topic

Relationship of multivariate normal to χ^2-distribution

Lemma 11.2. i) $\underset{\sim}{Z} \sim N_q(\underset{\sim}{0}, I) \Rightarrow \underset{\sim}{Z}'\underset{\sim}{Z} \sim \chi^2_q$.

ii) $\underset{\sim}{X} \sim N_q(\underset{\sim}{\mu}, \Sigma)$, $\Sigma > 0 \Rightarrow D^2_q(\underset{\sim}{X}) = (\underset{\sim}{X} - \underset{\sim}{\mu})'\Sigma^{-1}(\underset{\sim}{X} - \underset{\sim}{\mu}) \sim \chi^2_q$.

Proof: i) $\underset{\sim}{Z}'\underset{\sim}{Z} = \sum_{i=1}^{q} Z_i^2$ with $Z_1^2, \ldots, Z_q^2$ iid χ^2_1

$\sim \chi^2_q$

ii) $D^2_q(\underset{\sim}{X}) = \underset{\sim}{Z}'\underset{\sim}{Z}$ where $\underset{\sim}{Z} = \Sigma^{-\frac{1}{2}}(\underset{\sim}{X} - \underset{\sim}{\mu}) \sim N_q(\underset{\sim}{0}, I)$

$\sim \chi^2_q$. □

- What about singular Σ?

REVIEW: ORTHOGONAL PROJECTIONS IN EUCLIDEAN SPACE

DEF. 11.3. $Q_{q\times q}: \mathbb{R}^q \to \mathbb{R}^q$ is an orthogonal projection matrix, i.e. LINEAR OPERATOR, ONTO a k-dimensional subspace $\mathcal{V} \subset \mathbb{R}^q$ IFF

i) $Q\underset{\sim}{x} = \underset{\sim}{x}$ for all $\underset{\sim}{x} \in \mathcal{V}$.

ii) FOR ANY $\underset{\sim}{y} \in \mathbb{R}^q$, $\underset{\sim}{x} \in \mathcal{V}$, $\underset{\sim}{y} - \hat{\underset{\sim}{y}} \perp \underset{\sim}{x}$, i.e. $\underset{\sim}{x}'(\underset{\sim}{y} - \hat{\underset{\sim}{y}}) = 0$,

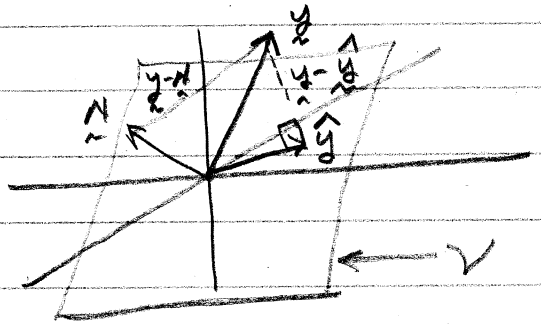
WHERE $\hat{\underset{\sim}{y}} = Q\underset{\sim}{y}$ is called the <u>orthogonal projection</u> of $\underset{\sim}{y}$ ONTO $\mathcal{V}$.

LEMMA 11.4.

a) THE LINEAR OPERATOR Q EXISTS AND IS UNIQUELY DEFINED, NAMELY, $Q = PP'$ WHERE $P = [\underset{\sim}{p}_1 \cdots \underset{\sim}{p}_k]$ IS an orthonormal basis for $\mathcal{V}$.

b) $\|\underset{\sim}{y} - \hat{\underset{\sim}{y}}\| \le \|\underset{\sim}{y} - \underset{\sim}{x}\|$, for any $\underset{\sim}{x} \in \mathcal{V}$, with equality IFF $\underset{\sim}{x} = \hat{\underset{\sim}{y}}$.

GRAPHICALLY



PROOF: a) i) $\underset{\sim}{x} \in \mathcal{V} \Rightarrow \underset{\sim}{x} = \sum_{j=1}^{k} a_j \underset{\sim}{p}_j$ and so $P'\underset{\sim}{x} = \sum_{j=1}^{k} a_j \underset{\sim}{e}_j$

where $\underset{\sim}{e}_j$ is jth EUCLIDEAN BASIS ELEMENT

$\Rightarrow Q\underset{\sim}{x} = PP'\underset{\sim}{x} = \sum_{j=1}^{k} a_j \underset{\sim}{p}_j = \underset{\sim}{x}$ HENCE: $Q\underset{\sim}{x} = \underset{\sim}{x}$.

ii) $\underset{\sim}{x} \in \mathcal{V}$, $\underset{\sim}{y} \in \mathbb{R}^q \Rightarrow \underset{\sim}{x}'(\underset{\sim}{y} - \hat{\underset{\sim}{y}}) = \underset{\sim}{x}'(I - PP')\underset{\sim}{y} = \underset{\sim}{x}'\underset{\sim}{y} - \underset{\sim}{x}'\hat{\underset{\sim}{y}} = 0$.

HENCE, $Q = PP'$ SATISFIES THE DEFINITION. TO SHOW IT UNIQUELY SATISFIES THE DEFINITION, FIRST PROVE (b). TO DO SO, EXPRESS

$$\|y - Nx\|^2 = \|y - \hat{y} + \hat{y} - Nx\|^2 = \|y - \hat{y}\|^2 + \|\hat{y} - Nx\|^2 \text{ SINCE}$$

THE CROSS PRODUCT TERM $2(y - \hat{y})'(\hat{y} - Nx) = 0$

THUS, $\|y - Nx\| \geq \|y - \hat{y}\|$ with equality ONLY IF $\|\hat{y} - Nx\| = 0$.

FOR UNIQUENESS, SUPPOSE $A_{q \times q}$ ALSO SATISFIES THE DEFINITION, then $\|y - Qy\| \leq \|y - Ay\| \leq \|y - Qy\| \Rightarrow \|y - Ay\| = \|y - Qy\|$

$\Rightarrow y'Ay = y'Qy$ all $y \in \mathbb{R}^q \Rightarrow A = Q$. (VERIFY) ∎

FURTHER USEFUL PROPERTIES. (HW: VERIFY)

1) $Q = PP'$ with $P'P = I_k$

2) $Q = Q^2$ AND $Q = Q'$. THAT IS Q IS SYMMETRIC AND IDEMPOTENT.

3) $V = \text{RANGE}(Q) = \text{RANGE}(P)$

4) EIGENVALUES OF Q: $\lambda = 1$ with multiplicity k
$\lambda = 0$ with multiplicity $(q - k)$.

5) $\text{TR}(Q) = k = \text{RANK}(Q)$.

6) ANY SYMMETRIC AND IDEMPOTENT MATRIX, say Q_0, CORRESPONDS TO AN ORTHOGONAL PROJECTION MATRIX ONTO ITS RANGE.

RETURNING TO QUADRATIC FORMS.

LEMMA 11.5. IF $\underset{\sim}{Z} \sim N_q(0, I)$ and $Q = Q^2 = Q'$ with RANK$(Q) = k$

THEN $\underset{\sim}{Z}'Q\underset{\sim}{Z} \sim \chi^2_k$

PROOF: EXPRESS $Q = PP'$, then $\underset{\sim}{Z}'Q\underset{\sim}{Z} = \underset{\sim}{Z}'_0\underset{\sim}{Z}_0 \sim \chi^2_k$ SINCE $\underset{\sim}{Z}_0 = P'\underset{\sim}{Z} \sim N_R(0, I)$

THEOREM 11.6. IF $\underset{\sim}{X} \sim N_q(\underset{\sim}{\mu}, \Sigma)$ with $\text{rank}(\Sigma) = k \leq q$,

THEN $(\underset{\sim}{X} - \underset{\sim}{\mu})'\Sigma^-(\underset{\sim}{X} - \underset{\sim}{\mu}) \sim \chi^2_k$ where Σ^- IS THE MOORE-PENROSE GENERALIZED INVERSE OF Σ.

DEF 11.7. IF $\underset{q\times q}{\Gamma}$ IS SYMMETRIC WITH SPECTRAL VALUE DECOMPOSITION $\Gamma = P\Delta P' = \sum_{j=1}^{q} \lambda_j \underset{\sim}{P}_j \underset{\sim}{P}_j'$, then THE MOORE-PENROSE g-INVERSE IS DEFINED TO BE $\Gamma^- = P\Delta^- P' = \sum_{j=1}^{q} \lambda_j^- \underset{\sim}{P}_j \underset{\sim}{P}_j'$ with $\lambda^- = \begin{cases} \frac{1}{\lambda}, & \lambda \neq 0 \\ 0, & \lambda = 0 \end{cases}$

PROOF OF THEOREM 11.6. FOR $\Sigma = P\Delta P'$; $\underset{\sim}{X} \sim \underset{\sim}{\mu} + P\Delta^{1/2}\underset{\sim}{Z}$

WHERE $\underset{\sim}{Z} \sim N_q(\underset{\sim}{0}, I) \Rightarrow (\underset{\sim}{X} - \underset{\sim}{\mu})'P\Delta^-P'(\underset{\sim}{X} - \underset{\sim}{\mu}) \sim \underset{\sim}{Z}'\Delta^{1/2}\Delta^-\Delta^{1/2}\underset{\sim}{Z}$

$\sim \underset{\sim}{Z}'Q\underset{\sim}{Z}$ where $Q = \Delta^{1/2}\Delta^-\Delta^{1/2} = \begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix}$

$\sim \chi^2_k$, SINCE Q IS AN ORTHOGONAL PROJECTION OF RANK k. ∎

NOTE: IF $\Sigma = AMA'$ WHERE A IS NONSINGULAR AND M IS SYMMETRIC, THEN $\Sigma^- \neq (A')^{-1} M^- A^{-1}$ IN GENERAL. (HW: FIND COUNTER-EXAMPLE)

HOWEVER, THEOREM 11.6 STILL HOLDS IF Σ^- IS REPLACED BY $(A')^{-1} M^- A^{-1}$. (HW: VERIFY)

REVIEW OF LIMIT THEOREMS IN $\mathbb{R}^q$

- A SEQUENCE OF RANDOM VECTORS $\underset{\sim}{X}_n \in \mathbb{R}^q$ CONVERGES IN DISTRIBUTION TO $\underset{\sim}{X} \in \mathbb{R}^q$ ($\underset{\sim}{X}_n \to_d \underset{\sim}{X}$) IFF $F_n(\underset{\sim}{x}) \to F(\underset{\sim}{x})$ at every CONTINUITY POINT OF $\underset{\sim}{x}$, where $\underset{\sim}{X}_n \sim F_n$ and $\underset{\sim}{X} \sim F$.

- EQUIVALENTLY, IFF $\phi_n(\underset{\sim}{t}) = E[e^{i \underset{\sim}{t}' \underset{\sim}{X}_n}] \to \phi(\underset{\sim}{t}) = E[e^{i \underset{\sim}{t}' \underset{\sim}{X}}]$ IN A NEIGHBORHOOD OF $\underset{\sim}{t} = \underset{\sim}{0}$.

- EQUIVALENTLY, IFF $P(\underset{\sim}{X}_n \in B) \to P(\underset{\sim}{X} \in B)$ EVERY OPEN BOREL SET $B \subset \mathbb{R}^q$.

LEMMA 11.8. i) $\underset{\sim}{X}_n \to_d \underset{\sim}{X}$, $g: \mathbb{R}^q \to \mathbb{R}^p$, CONTINUOUS $\Rightarrow g(\underset{\sim}{X}_n) \to_d g(\underset{\sim}{X})$

ii) $\underset{\sim}{X}_n \to_p \underset{\sim}{X} \iff \underset{\sim}{X}_n - \underset{\sim}{X} \to_p \underset{\sim}{0} \iff \underset{\sim}{X}_n - \underset{\sim}{X} \to_d \underset{\sim}{0}$

IN GENERAL, $\underset{\sim}{X}_n \to_p \underset{\sim}{c}$, A CONSTANT VECTOR $\iff \underset{\sim}{X}_n \to_d \underset{\sim}{c}$.

NOTE: $\underset{\sim}{X}_n \to_d \underset{\sim}{X} \not\Rightarrow \underset{\sim}{X}_n - \underset{\sim}{X} \to \underset{\sim}{0}$

iii) $\underset{\sim}{X}_n \to_{as} \underset{\sim}{X} \iff \underset{\sim}{X}_n - \underset{\sim}{X} \to_{as} \underset{\sim}{0}$.

LEMMA 11.9.. $\underset{\sim}{X}_n \xrightarrow{d} \underset{\sim}{X} \iff \underset{\sim}{a}' \underset{\sim}{X}_n \xrightarrow{d} \underset{\sim}{a}' \underset{\sim}{X}$ for all $\underset{\sim}{a} \in \mathbb{R}^q$.

PROOF: VIA CHARACTERISTIC FUNCTIONS (HW)

MULTIVARIATE WLLN AND CLT.

IF $\underset{\sim}{X}_1, \ldots, \underset{\sim}{X}_n$ iid $\underset{\sim}{X}$ with $E(\underset{\sim}{X}) = \underset{\sim}{\mu}$ EXISTING, then

i) $\bar{\underset{\sim}{X}} \xrightarrow{p} \underset{\sim}{\mu}$, AND

ii) $\sqrt{n}(\bar{\underset{\sim}{X}} - \underset{\sim}{\mu}) \xrightarrow{d} N_q(\underset{\sim}{0}, \underset{\sim}{\Sigma})$ PROVIDED $\mathrm{VAR}(\underset{\sim}{X}) = \underset{\sim}{\Sigma}$ EXISTS.

PROOF: APPLICATION OF LEMMA 11.9. (HW)

LEMMA 11.10. FOR A SEQUENCE OF ESTIMATION PROBLEMS,

i) SUPPOSE $\hat{\underset{\sim}{\theta}}_n$ IS AN ESTIMATE OF $\underset{\sim}{\theta}$ SUCH THAT

$$\sqrt{n}(\hat{\underset{\sim}{\theta}}_n - \underset{\sim}{\theta}) \to N_q(\underset{\sim}{0}, \underset{\sim}{\Sigma}_\theta) \quad \text{IN DISTRIBUTION}$$

with $\mathrm{rank}(\underset{\sim}{\Sigma}_\theta) = k$, then

$$T_n = n(\hat{\underset{\sim}{\theta}}_n - \underset{\sim}{\theta})' \underset{\sim}{\Sigma}_\theta^{-} (\hat{\underset{\sim}{\theta}}_n - \underset{\sim}{\theta}) \xrightarrow{d} \chi^2_k$$

ii) FURTHERMORE, SUPPOSE $\hat{\underset{\sim}{\Sigma}}_n \to \underset{\sim}{\Sigma}_\theta$ IN PROBABILITY, WITH

$P_\theta(\mathrm{RANK}(\hat{\underset{\sim}{\Sigma}}_n) = k) \to 1$, then

$$T_n^* = n(\hat{\underset{\sim}{\theta}}_n - \underset{\sim}{\theta})' \hat{\underset{\sim}{\Sigma}}_n^{-} (\hat{\underset{\sim}{\theta}}_n - \underset{\sim}{\theta}) \xrightarrow{d} \chi^2_k .$$

PROOF:

i) $T_n = \text{TRACE}\left(\Sigma_\theta^- \underset{\sim}{X}_n \underset{\sim}{X}_n'\right)$ where $\underset{\sim}{X}_n = \sqrt{n}(\hat{\underset{\sim}{\theta}}_n - \underset{\sim}{\theta}) \to_d \underset{\sim}{X} \sim N(\underset{\sim}{0}, \Sigma_\theta)$

SO BY CONTINUITY, $T_n \to_d T = \text{TRACE}\left(\Sigma_\theta^- \underset{\sim}{X}\underset{\sim}{X}'\right) = \underset{\sim}{X}'\Sigma_\theta^- \underset{\sim}{X}$

with $T \sim \chi^2_k$ by THEOREM 11.6.

ii) $T_n^* = \text{TRACE}\left(\hat{\Sigma}_n^- \underset{\sim}{X}_n \underset{\sim}{X}_n'\right) \to_d \text{TRACE}\left(\Sigma_\theta^- \underset{\sim}{X}\underset{\sim}{X}'\right) \sim \chi^2_k$

by SLUTSKY'S THEOREM (SPECIAL CASE OF 11.8)

SINCE THE FOLLOWING LEMMA HOLDS, HENCE $\hat{\Sigma}_n^- \to_p \Sigma_\theta^-$.

LEMMA 11.11. IF $\Sigma_k \to \Sigma$ and $\text{RANK}(\Sigma_k) \to \text{RANK}(\Sigma)$, then $\Sigma_k^- \to \Sigma^-$. THAT IS, MOORE-PENROSE g-INVERSE IS CONTINUOUS AT POINTS WHERE THE RANK IS CONTINUOUS.

PROOF: ADVANCED, REQUIRES CONTINUITY PROPERTIES OF S.V.D.

NOTE: $\text{RANK}(\Sigma_k) \to \text{RANK}(\Sigma) \Rightarrow \text{RANK}(\Sigma_k) = \text{RANK}(\Sigma)$ LARGE ENOUGH k.

NOTE: IF $\Sigma_\theta = AMA'$, A NONSINGULAR, THEN THEOREM 13.10 i STILL HOLDS IF Σ_θ^- IS REPLACED BY $(A')^{-1}M^-A^-$.

° ANALOGOUS STATEMENT HOLDS FOR 11.10.ii.

NEXT TIME: APPLICATION TO WALD'S χ^2-TESTS AND ELLIPTICAL CONFIDENCE REGIONS.